

Analysis First-Year Exam, May 2025, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. The sequence $(f_n)_{n=0}^\infty$ of real-valued functions on $[0, 1]$ is uniformly bounded and converges to 0 pointwise. If each f_n is (i) continuous, (ii) Riemann integrable, (iii) Lebesgue integrable, does it follow that $\int_0^1 f_n(t) dt \rightarrow 0$? In each of the three cases, a proof or counterexample should be given.
2. Let $(f_n)_{n=0}^\infty$ be a sequence of continuous non-negative functions on $[0, 1]$ converging pointwise to 0. Prove that if the extra condition

$$\forall t \in [0, 1] \quad f_0(t) \geq f_1(t) \geq \dots \geq f_n(t) \geq \dots$$

is satisfied then $f_n \rightarrow 0$ uniformly on $[0, 1]$. Show that this conclusion can fail if the extra condition is removed.

3. Let the continuous real-valued function f on $[0, 1]$ satisfy

$$\forall n \in \mathbb{N} \quad \int_0^1 f(t)(1-t)^n dt = 0.$$

Deduce as much as possible about f .

4. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. Let (A_n) be a sequence in $\mathcal{A}$ and define

$$\liminf A_n = \bigcup_N \bigcap_{n \geq N} A_n.$$

Prove that

$$\mu(\liminf A_n) \leq \liminf \mu(A_n).$$

5. Let $f(x) = \sin x/x$ for $x > 0$. Show that if n is a positive integer then

$$\int_{2n\pi}^{(2n+1)\pi} \frac{\sin x}{x} dx \geq \frac{2}{(2n+1)\pi}.$$

Hence explain why f is not Lebesgue integrable on (a, ∞) for any $a > 0$.